

Predicting MLB Draft Outcomes Using a Bayesian Two-Part Hierarchical Model

Evan Delcamp
University of Michigan
Ann Arbor, MI, USA
edelcamp@umich.edu

Abstract—This paper presents a Bayesian two-part hierarchical model that separates draft outcomes into (i) the probability of reaching MLB within seven years, and (ii) how productive a player is, conditional on reaching MLB. This model is based on a decomposition of pre-draft covariates (e.g., age, school type, and draft ranking signals) and provides player-level uncertainty intervals that can be used for risk-aware comparisons. Across analyses, published ranking signals were strongly associated with higher reach probabilities and higher unconditional expected WAR, and group-level posterior summaries showed that ranked subgroups dominate expected value across school types and roles.

Index Terms—Bayesian inference, hierarchical models, hurdle model, Stan, baseball analytics, MLB draft

I. INTRODUCTION

The Major League Baseball (MLB) draft has a lot of variation. Players drafted in the 4th or 5th round or later can end up being in the Hall of Fame, and some first overall picks never play a single game at the highest level. In this paper, we attempt to see the feasibility of using covariates known pre-draft to understand player successes, using Bayesian methods to produce posterior player level distributions describing player success probabilities and uncertainty intervals as a measure of individual player risk. A literature review of public research of this approach for the MLB amateur draft did not return any results, though it would seem likely that MLB teams have their own Bayesian player models in some capacity.

A common metric used to roughly quantify player contribution toward team wins is Wins Above Replacement (WAR). WAR is a single-number summary of total player contribution; roughly, in one season, 2–3 WAR is an everyday regular and 4+ WAR is All-Star level [0].

WAR for drafted players is a statistic with a long tail; much of the average is pulled up by the few star level players. Some teams may want maximize drafted players who will reach MLB, and some may prefer highest upside. For this reason, a posterior distribution for each player that gives an upside factor (uncertainty) would be useful. Therefore, this paper attempts to understand the two distinct components that determine a draftee’s eventual value: (i) whether the player reaches MLB at all, and (ii) how productive the player is, conditional on reaching MLB. Specifically, this paper studies how pre-draft covariates (e.g., age, school type, draft ranking signals) relate to both components of draft “success” using a Bayesian two-part hierarchical model fit to the 2017–2018

draft classes. To keep modeling simple, and assuming that team’s would prefer to draft players that produce at the MLB level sooner rather than later, we will use a 7 full year cutoff window for a player to reach MLB and produce WAR. This decomposition yields interpretable covariate effects on each stage and provides player-level uncertainty intervals that can be used for risk-aware comparisons.

II. DATASET

We analyze $N = 2360$ signed draftees from the 2017–2018 MLB drafts assembled from `baseballr`. Outcomes are (i) reaching MLB within 7 years and (ii) 7-year WAR among reachers; predictors include age, school type, pitcher indicator, handedness, and a pre-draft ranking indicator/score. For one player, the dataset is as described in Table II, including information about necessary transformations for regression. Some brief summary statistics are in Table III.

TABLE I
OUTCOMES AND PREDICTORS USED IN THE MODEL

Variable	Description / coding	Used in
<code>reached_mlb_7yr</code>	Indicator for reaching MLB within 7 years (0/1).	Reach
<code>WAR_7yr</code>	Seven-year Wins Above Replacement; modeled only for players with <code>reached_mlb_7yr=1</code> .	WAR
<code>age_june_1</code>	Age (months) as of June 1 of draft year; centered for modeling as $age_c = age_june_1 - age_june_1$.	Both
<code>school_type_3c</code>	Categorical: High School / College / JUCO.	Both
<code>is_pitcher</code>	Indicator for primary position = pitcher (0/1).	Both
<code>bat_side</code>	Categorical: R / L / S.	Both
<code>throw_hand</code>	Categorical: R / L.	Both
<code>ranked</code>	Indicator for presence on pre-draft ranking list (0/1).	Both
<code>rank_score</code>	If ranked, $201 - rank + 1$, else 0.	Both
<code>year_id</code>	Draft-year index used for hierarchical intercepts (2017–2018).	Both

This dataset excludes unsigned draftees, so results should be interpreted as conditional on signable/signed players, not all drafted players.

III. METHOD

Exploratory frequentist regressions suggested that pre-draft ranking signals are strong predictors of reaching MLB, while

TABLE II

SUMMARY STATISTICS FOR THE ANALYTIC SAMPLE (2017–2018 DRAFTS)

Statistic	Value
Players (N)	2360
Draft years	2017–2018
2017 count	1146
2018 count	1214
Reached MLB within 7 years	427 (18.1%)
Pitchers ($\text{is_pitcher}=1$)	1284 (54.4%)
Ranked prospects ($\text{ranked}=1$)	377 (16.0%)
School type: High School	615 (26.1%)
School type: College	1586 (67.2%)
School type: JUCO	159 (6.7%)
Age at draft (years), mean (SD)	20.84 (1.66)
WAR ₇ (all players), mean	0.24
WAR ₇ reach, mean (SD)	1.35 (3.08)
WAR ₇ reach, median [IQR]	0.19 [−0.16, 1.46]

showing a weaker association with WAR|reach. This motivates a Bayesian two-part model that separately (i) models the probability of reaching MLB within seven years via logistic regression and (ii) models seven-year WAR among players who reached MLB using a regularized Normal regression. This two-stage structure matches the observed outcome pattern: many players have zero WAR because they never reach MLB, while players who do reach MLB exhibit a continuous distribution of WAR values.

Baseline rates of reaching MLB and baseline productivity can vary by draft year. To capture this, both model components include year-specific intercepts (baselines) for a reference player ($x_i = 0$). These year intercepts are modeled hierarchically (partially pooled), which shrinks extreme year effects toward an overall mean and allows the model to learn the magnitude of between-year variation.

Although the current dataset contains only two draft years (2017 and 2018), year-specific intercepts were kept as part of the model to account for baseline differences across years and to support extension to additional draft years in future work.

A. Model overview

Let $z_i \in \{0, 1\}$ denote whether player i reached MLB within seven years, and let y_i denote seven-year WAR (modeled only for players with $z_i = 1$). The model is

$$z_i \sim \text{Bernoulli}(p_i), \quad (1)$$

$$\text{logit}(p_i) = \alpha_{t(i)}^{(Z)} + x_i^\top \beta_Z, \quad (2)$$

and conditional on reaching MLB,

$$y_i | (z_i = 1) \sim \mathcal{N}(\mu_i, \sigma_Y), \quad (3)$$

$$\mu_i = \alpha_{t(i)}^{(Y)} + x_i^\top \beta_Y. \quad (4)$$

B. Hierarchical year effects (partial pooling)

For the year-specific intercepts in the reach component,

$$\alpha_t^{(Z)} = \mu_{\alpha Z} + \tau_{\alpha Z} \alpha_{\text{raw},t}^{(Z)}, \quad (5)$$

$$\alpha_{\text{raw},t}^{(Z)} \sim \text{Normal}(0, 1), \quad (6)$$

TABLE III
MODEL NOTATION

Symbol	Definition
i	Player index, $i = 1, \dots, N$.
$t(i)$	Draft-year index for player i , $t(i) \in \{1, \dots, T\}$.
z_i	Indicator that player i reached MLB within 7 years (1 if yes, 0 otherwise).
y_i	Seven-year WAR for player i (modeled only for players with $z_i = 1$).
x_i	Vector of pre-draft covariates for player i (row of the design matrix X).
p_i	Reach probability, $p_i = P(z_i = 1 x_i, t(i))$.
β_Z	Regression coefficients for the reach (logistic) component.
β_Y	Regression coefficients for the WAR reach component.
$\alpha_t^{(Z)}$	Draft-year intercept for reach probability (log-odds scale).
$\alpha_t^{(Y)}$	Draft-year intercept for WAR reach (WAR scale).
$\mu_{\alpha Z}, \tau_{\alpha Z}$	Hyperparameters controlling the mean and SD of $\alpha_t^{(Z)}$.
$\mu_{\alpha Y}, \tau_{\alpha Y}$	Hyperparameters controlling the mean and SD of $\alpha_t^{(Y)}$.
σ_Y	Residual SD in the WAR reach model.

and similarly for the conditional WAR component,

$$\alpha_t^{(Y)} = \mu_{\alpha Y} + \tau_{\alpha Y} \alpha_{\text{raw},t}^{(Y)}, \quad (7)$$

$$\alpha_{\text{raw},t}^{(Y)} \sim \text{Normal}(0, 1). \quad (8)$$

In these, $\alpha_{\text{raw},t}^{(\cdot)}$ denotes a standardized (“raw”) year effect, and $\alpha_t^{(\cdot)} = \mu_{\alpha(\cdot)} + \tau_{\alpha(\cdot)} \alpha_{\text{raw},t}^{(\cdot)}$ is the corresponding non-centered parameterization used for improved MCMC.

C. Prior selection

We use weakly informative priors to regularize estimation while keeping effect sizes in plausible ranges.

a) *Reach (logistic) coefficients*: For the reach model, coefficients are on the log-odds scale: we use $\beta_Z \sim \text{Normal}(0, 1.5)$ to allow some (but not extreme) shifts in odds.

b) *WAR conditional on reaching*: Exploratory analysis suggested that pre-draft covariates explain only modest variation in WAR among MLB reachers, so we impose stronger shrinkage with $\beta_Y \sim \text{Normal}(0, 0.5)$, expressing skepticism that a one-unit covariate change moves expected seven-year WAR by multiple wins.

c) *Residual variation*: We model remaining heterogeneity in WAR among reachers via $\sigma_Y \sim \text{Exponential}(0.5)$ (mean 2), which mildly favors small residual noise but allows values in the realistic range observed for seven-year WAR.

d) *Draft-year intercepts (partial pooling)*: Year intercepts are modeled hierarchically to capture baseline differences across draft years: $\alpha_t^{(Z)} = \mu_{\alpha Z} + \tau_{\alpha Z} \alpha_{\text{raw},t}^{(Z)}$ and $\alpha_t^{(Y)} = \mu_{\alpha Y} + \tau_{\alpha Y} \alpha_{\text{raw},t}^{(Y)}$ with $\alpha_{\text{raw},t}^{(\cdot)} \sim \mathcal{N}(0, 1)$, $\mu_{\alpha Z} \sim \mathcal{N}(0, 1)$, $\mu_{\alpha Y} \sim \mathcal{N}(0, 2)$, and $\tau_{\alpha Z}, \tau_{\alpha Y} \sim \text{Exponential}(1)$. The wider prior on $\mu_{\alpha Y}$ reflects the larger scale of WAR relative to log-odds, allowing typical year-level conditional WAR among MLB reachers to plausibly vary by several wins a priori.

D. Computation and posterior quantities

The model was fit in Stan via `rstan` using 4 chains of 2000 iterations each (1000 warmup), with

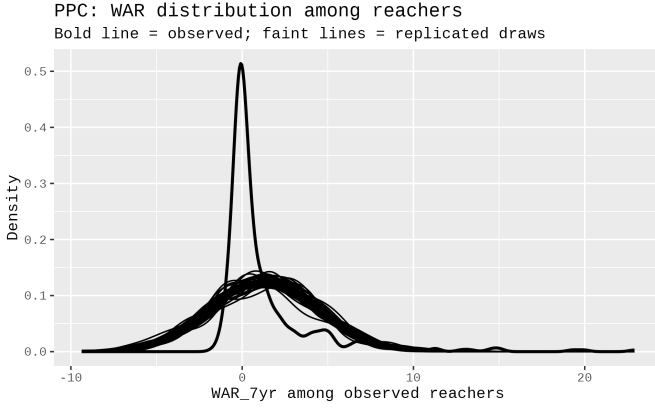


Fig. 1. Posterior predictive check for the WAR|reach component. Bold curve: observed distribution of seven-year WAR among players who reached MLB. Faint curves: replicated datasets drawn from fitted model. The Normal likelihood reproduces scale but does not capture pile-up near 0 WAR.

$\text{adapt_delta}=0.99$ and $\text{max_treedepth}=12$. With only $T = 2$ years, the year-effect scale parameters are weakly identified, which led to early MCMC runs having divergent transitions; increasing adapt_delta to 0.99 yielded stable posterior sampling. This also led to $\tau_{\alpha Z}$ exhibiting a right-skewed posterior and slightly higher $\hat{R}$ (1.01). Ultimately, all reported parameters had $\hat{R} \approx 1.00$ with r_{eff} sizes in the thousands. There were 0 divergent transitions, 0 iterations hitting the maximum tree depth (12) suggesting stable sampling.

From posterior draws, we compute the following player-level quantities. First, the posterior reach probability

$$p_i = \text{logit}^{-1} \left(\alpha_{t(i)}^{(Z)} + x_i^\top \beta_Z \right),$$

and the posterior conditional mean WAR among reachers

$$\mu_i = \alpha_{t(i)}^{(Y)} + x_i^\top \beta_Y.$$

For each player, we report posterior means and 90% credible intervals for p_i and μ_i . Group-level summaries are obtained by averaging the player-level posterior means within subgroups (e.g. ranked vs. unranked).

E. Model checking

Figure 1 presents a posterior predictive check for the WAR|reach component. Among MLB reachers, seven-year WAR is highly concentrated near zero (about 60% are below 0.5 WAR) with a long right tail (95th percentile ≈ 7.2 WAR). The fitted Normal likelihood reproduces the broad scale of WAR but underestimates the near-zero pile-up. A quantile-based PPC is consistent with this mismatch: observed median WAR among reachers is 0.19, but replicated datasets place the median between 0.94 and 1.73, and the model produces an unrealistically negative lower tail.

Although the Normal likelihood is not flexible enough to reproduce the empirical shape of WAR | reach, the model can still be used for player-level regression summaries because the MCMC output provides a posterior distribution over the global parameters (e.g., $\beta_Z, \beta_Y, \alpha^{(Z)}, \alpha^{(Y)}, \sigma_Y$) and this

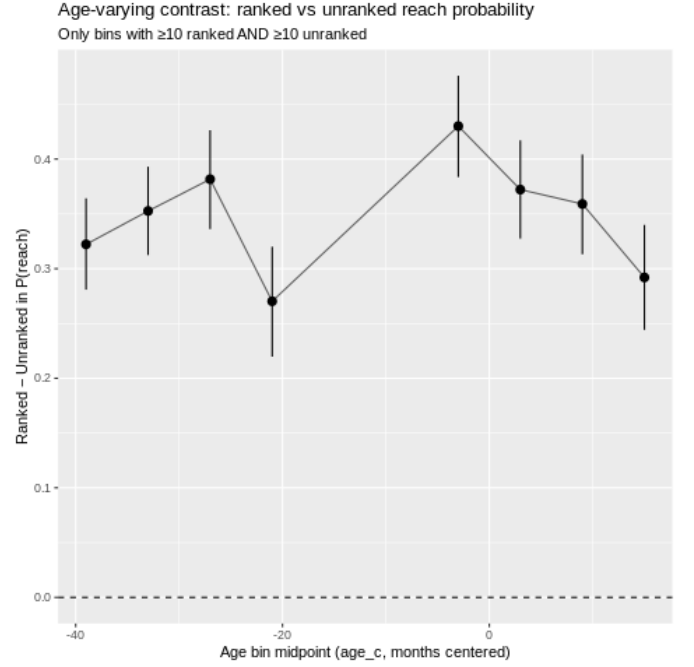


Fig. 2. Ranking advantage by age bin: posterior median and 90% credible interval for $\Delta_b = \mathbb{E}[p_i | \text{ranked}, b] - \mathbb{E}[p_i | \text{unranked}, b]$. Bins with < 10 players per group omitted.

posterior induces a posterior distribution over each player's reach probability and conditional mean WAR through their covariates x_i . These quantities describe how covariates shift the expected outcome in each stage, with uncertainty intervals that reflect parameter uncertainty. Therefore, we will focus on these player-level outcomes for results.

IV. RESULTS

A. Effect of Published Ranking for Players

A published ranking signal seems to be associated with players reaching MLB in 7 years. To expand, using the posterior, we can answer whether that holds across all age groups. Across all age bins with sufficient sample size, the model assigns higher reach probabilities to players with a published ranking signal than to unranked players. Figure 2 plots the posterior distribution of the contrast $\Delta_b = \mathbb{E}[p_i | \text{ranked}, b] - \mathbb{E}[p_i | \text{unranked}, b]$ within age bin b . The contrast is consistently positive, indicating association between ranking and reaching MLB within seven years.

The Bayesian approach also provides a direct probabilistic interpretation: given the model and data, a 90% credible interval for Δ_b contains 90% of the posterior mass for the ranked–unranked difference in reach probability in age bin b .

B. Reach Probabilities by Player

Using draft-time covariates and a draft-year intercept, the model estimates player i 's probability of reaching MLB within seven years as $\hat{p}_i$ (posterior mean), with a 90% credible interval $[\hat{p}_{i,0.05}, \hat{p}_{i,0.95}]$. Figure 3 summarizes the players with the highest posterior reach probabilities. This is directly actionable

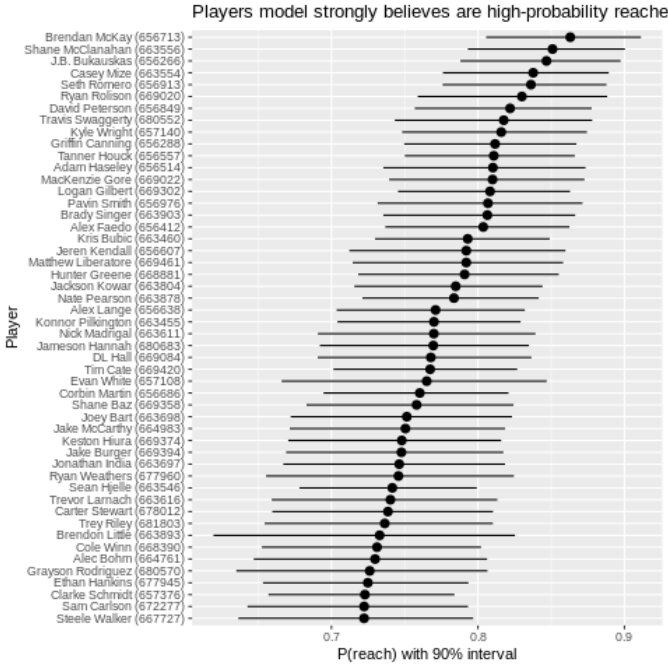


Fig. 3. Top Players for which the posterior strongly believes are high probability candidates to reach MLB.

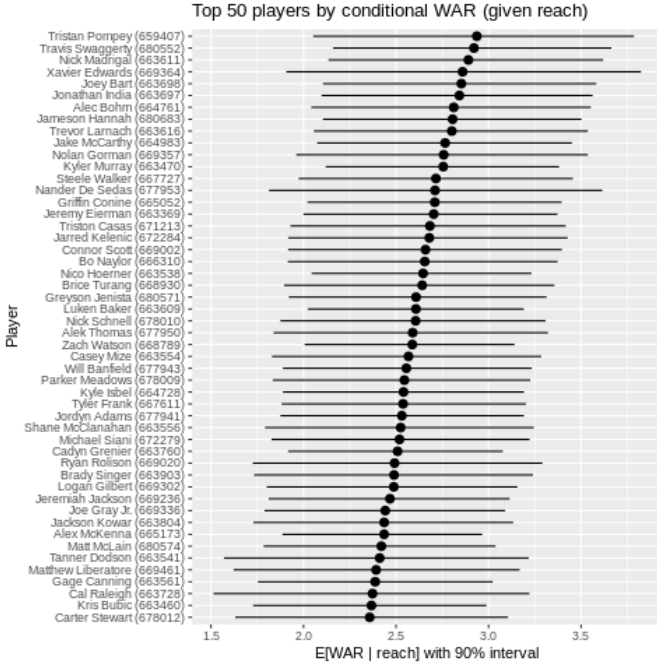


Fig. 4. Top 50 WAR Production — Reach

for a Major League Team looking to draft players likely to reach MLB and the biggest motivation for the model, along with conditional WAR among MLB reachers.

C. Conditional WAR among MLB reachers

Among players who reach MLB, the model estimates a conditional mean seven-year WAR, $\mu_i = E[\text{WAR}_i \mid \text{reach}]$,

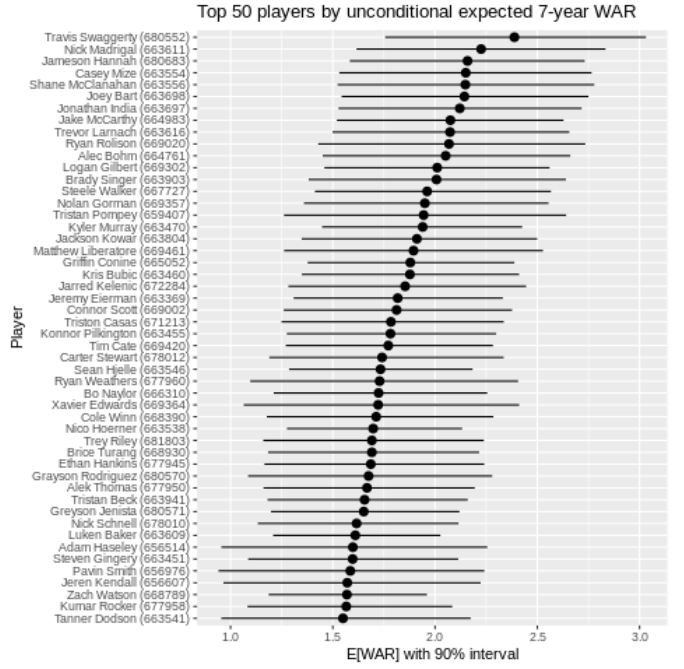


Fig. 5. Bottom 50 players by $E[\text{WAR} \mid \text{reach}]$ with 90% credible intervals. (Coincidentally, the lowest estimate in this subset was for a player named Askew.)

with uncertainty summarized by posterior intervals. The set of players with the largest μ_i differs from the set with the largest $\hat{p}_i$, illustrating the two-part structure: some players are modeled as highly likely to reach MLB but with only modest conditional productivity, while others have lower reach probability but higher expected WAR given reaching (Figure 4).

Posterior inference for the residual standard deviation concentrates around $\sigma_Y \approx 3$, indicating substantial variability in realized WAR even when conditional mean WAR is high.

This is also actionable for an MLB team, especially for a team looking for “upside” in the draft.

D. Unconditional expected value

To summarize total expected value on a common scale, we compute the unconditional expected seven-year WAR,

$$E[\text{WAR}_i] = P(\text{reach}_i) E[\text{WAR}_i \mid \text{reach}_i] = p_i \mu_i.$$

This quantity combines both stages of the model and provides a single expected-value accounting for both probability of reach and expected productivity once there (Figure 5).

Figure 6 shows posterior distributions of the *group-average* unconditional expected seven-year WAR, where each posterior draw averages $E[\text{WAR}_{7,i}]$ across players within a subgroup (school type $\times$ pitcher status $\times$ ranked status). Across all school categories and both pitcher/hitter roles, ranked players have substantially higher posterior group-average $E[\text{WAR}_7]$ than unranked players, indicating that ranking information is the dominant driver of expected total value in this model. Differences by school type and pitcher status are smaller than the ranked–unranked separation; in particular, ranked hitters

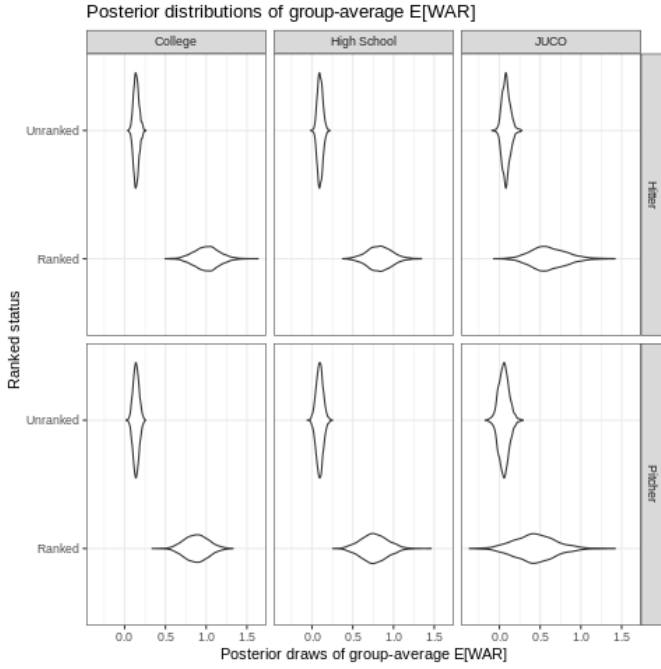


Fig. 6. Posterior distributions of group-average $E[\text{WAR}_7]$ by school type, pitcher status, and ranked status.

tend to have higher group-average expected WAR than ranked pitchers.

V. LIMITATIONS AND FUTURE WORK

This analysis is limited by both data availability and model specification. First, the dataset includes only two draft years (2017–2018), so the hierarchical year-effect scale parameters ($\tau_{\alpha Z}, \tau_{\alpha Y}$) are weakly identified and may not generalize to future draft years. Extending the dataset to more years would allow the model to better estimate between-year variability.

Second, posterior predictive checks indicate that the Normal likelihood for $\text{WAR}_7 \mid \text{reach}$ does not reproduce the empirical distribution among MLB reachers, which contains a pronounced spike near zero along with a long right tail. Unfortunately, the two parts did not correct for this, and this limits the realism of fully simulated WAR outcomes for new draft classes. A natural extension is to replace the single Normal likelihood with a more flexible model, such as a mixture model (e.g., a near-zero component plus a wider continuous component) or a heavier-tailed alternative.

Limited or missing predictors was also a limitation. Positions were simplified to pitcher vs. non-pitcher, and richer defensive/positional information could improve both components. College performance measures were not included because of difficulties in reliably linking player identifiers across data sources; incorporating amateur statistics would likely strengthen predictive accuracy. Finally, published ranking information is only available beginning in 2017, constraining the time horizon over which the model can be trained.

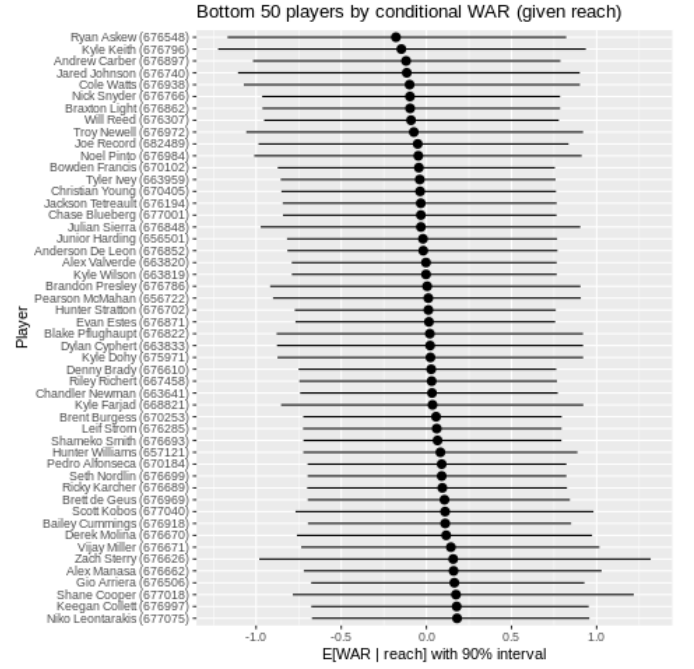


Fig. 7. Bottom 50 players by $E[\text{WAR} \mid \text{reach}]$ with 90% credible intervals. (Coincidentally, the lowest estimate in this subset was for a player named Askew.)

VI. CONCLUSION

We developed a Bayesian two-part hierarchical model that separates draft outcomes into (i) the probability of reaching MLB within seven years and (ii) seven-year WAR conditional on reaching. The model produces player-level posterior summaries for $P(\text{reach})$, $E[\text{WAR} \mid \text{reach}]$, and unconditional expected value $E[\text{WAR}] = p\mu$, enabling uncertainty-aware comparisons rather than point-estimate rankings.

Across analyses, published ranking signals were strongly associated with higher reach probabilities and higher unconditional expected WAR, and group-level posterior summaries showed that ranked subgroups dominate expected value across school types and roles. Posterior predictive checks highlighted a key limitation: a single Normal likelihood does not reproduce the observed $\text{WAR} \mid \text{reach}$ distribution, suggesting that more flexible likelihoods are needed for realistic forward simulation of WAR outcomes in new draft classes.

Overall, this project demonstrated how Bayesian inference produces player- and subgroup-level estimates with clear uncertainty intervals, and emphasizes the importance of model checking to validate a generative interpretation of fitted models.

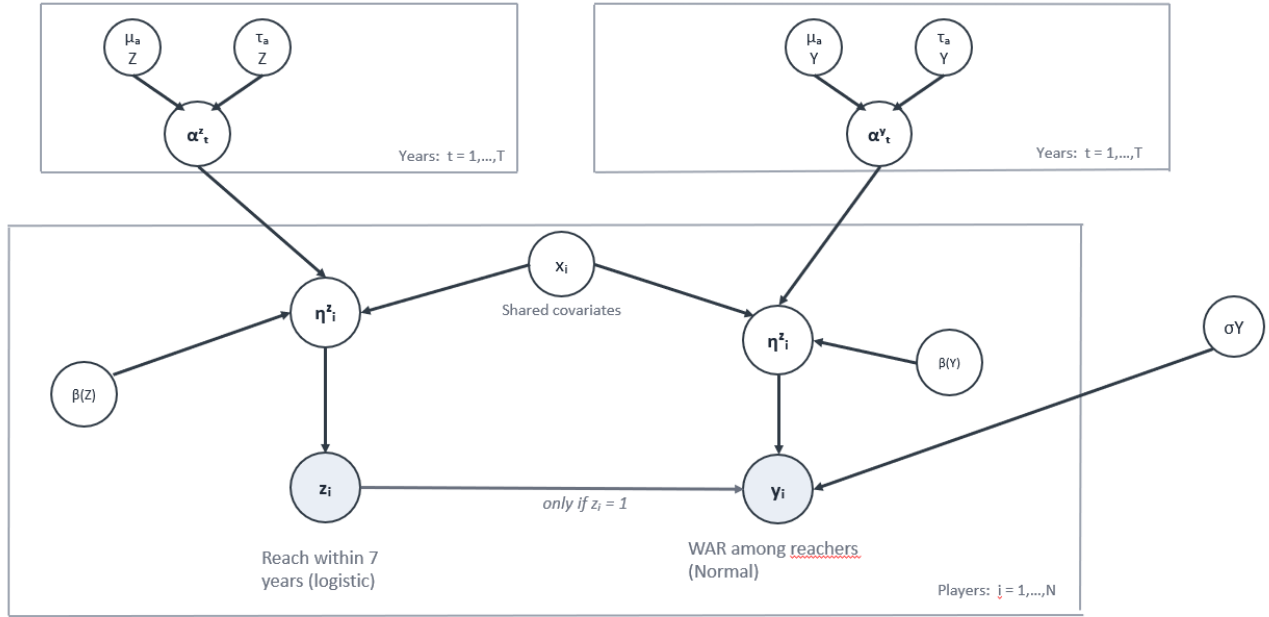
ACKNOWLEDGMENT

P. Slowinski, “What is WAR?,” *FanGraphs Sabermetrics Library*, Feb. 15, 2010. Accessed: Dec. 14, 2025. [Online]. Available: <https://library.fangraphs.com/misc/war/>

APPENDIX A FIGURES

Bayesian two-part hierarchical model (graphical)

Reach model: $z_i \sim \text{Bernoulli}(\text{logit}^{-1}(\eta_i^z))$ | WAR model: $y_i | z_i=1 \sim \text{Normal}(\mu_i, \sigma_Y)$



α_t^z and α_t^y are partially pooled year intercepts; x_i are pre-draft covariates; β_Z/β_Y are global coefficients.

Fig. 8. Bayesian graphical model for the conditional WAR component. A draft-year intercept α_t is drawn from a hierarchical prior with hyperparameters $(\mu_{\alpha}, \tau_{\alpha})$. For player i , covariates x_i and coefficients β form the linear predictor $\eta_i = \alpha_{t(i)} + x_i^T \beta$, and the observed outcome follows $y_i \sim \mathcal{N}(\eta_i, \sigma)$.